



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab Experiment – 08

Depth-First Search (DFS)

Aim

To implement the Depth-First Search (DFS) algorithm for graph traversal.

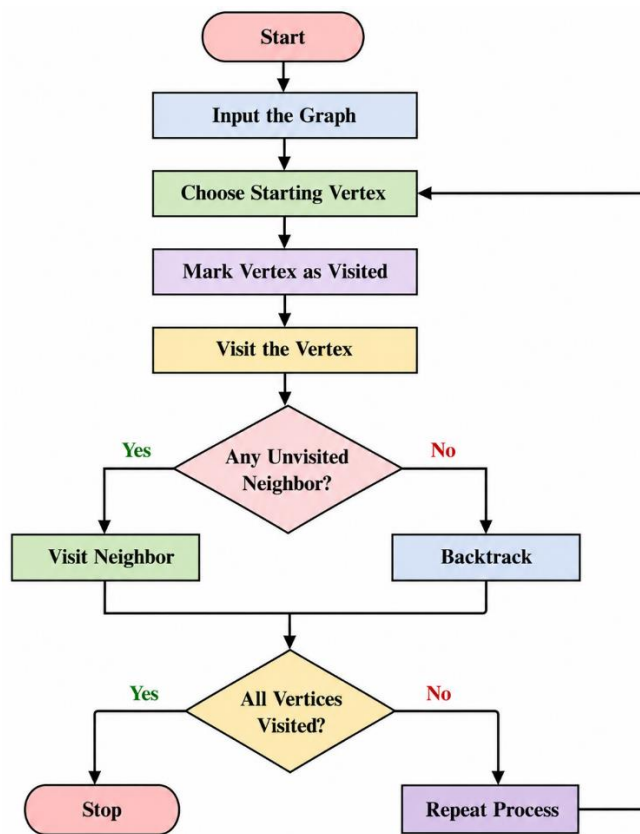
Objective

To traverse all the vertices of a graph using the Depth-First Search (DFS) algorithm.

Algorithm

1. Start with the source (starting) vertex.
2. Mark the starting vertex as visited.
3. Visit the current vertex.
4. Select an unvisited adjacent vertex.
5. If an unvisited adjacent vertex exists, recursively visit it.
6. If no unvisited adjacent vertex exists, backtrack to the previous vertex.
7. Repeat the process until all vertices have been visited.
8. Stop.

FlowChart



Code

```
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],  
    'C': ['F'],  
    'D': [],  
    'E': ['F'],  
    'F': []  
}
```

```
visited = set()
```

```
def dfs(vertex):
```

```
if vertex not in visited:
```

```
    print(vertex, end=" ")
```

```
    visited.add(vertex)
```

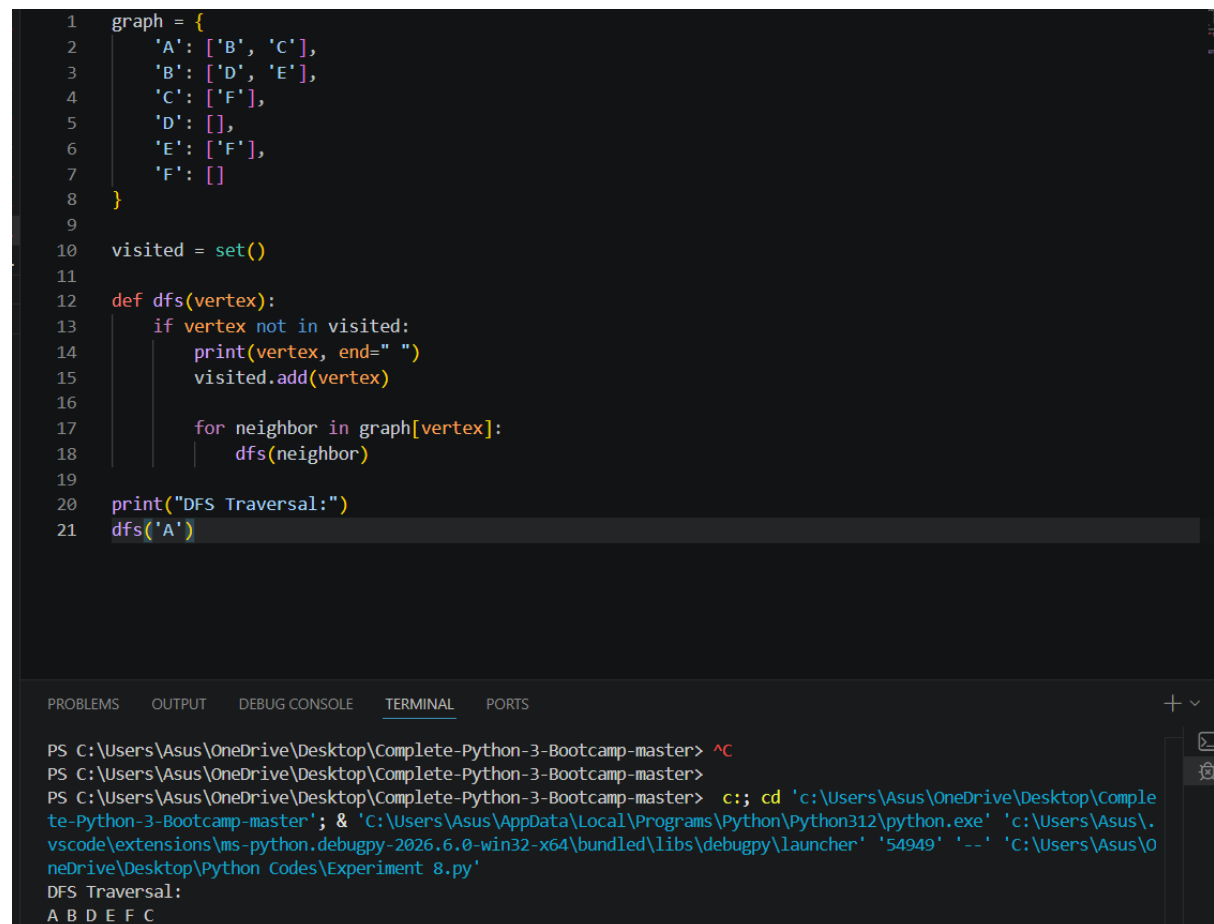
```
    for neighbor in graph[vertex]:
```

```
        dfs(neighbor)
```

```
print("DFS Traversal:")
```

```
dfs('A')
```

Output



```
1 graph = {
2     'A': ['B', 'C'],
3     'B': ['D', 'E'],
4     'C': ['F'],
5     'D': [],
6     'E': ['F'],
7     'F': []
8 }
9
10 visited = set()
11
12 def dfs(vertex):
13     if vertex not in visited:
14         print(vertex, end=" ")
15         visited.add(vertex)
16
17         for neighbor in graph[vertex]:
18             dfs(neighbor)
19
20 print("DFS Traversal:")
21 dfs('A')
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Asus\OneDrive\Desktop\Complete-Python-3-Bootcamp-master> ^C
PS C:\Users\Asus\OneDrive\Desktop\Complete-Python-3-Bootcamp-master>
PS C:\Users\Asus\OneDrive\Desktop\Complete-Python-3-Bootcamp-master> c::; cd 'c:\Users\Asus\OneDrive\Desktop\Complete-Python-3-Bootcamp-master'; & 'C:\Users\Asus\AppData\Local\Programs\Python\Python312\python.exe' 'c:\Users\Asus\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '54949' '--' 'C:\Users\Asus\OneDrive\Desktop\Python Codes\Experiment 8.py'
DFS Traversal:
A B D E F C
```

Result

The Depth-First Search (DFS) algorithm was implemented successfully, and all the vertices of the graph were traversed using depth-first traversal.

Conclusion

The DFS algorithm visits a graph by exploring as far as possible along each branch before backtracking. It uses recursion (or a stack) and is useful for graph traversal, path finding, and solving search problems.